

Homework 1

CHAPTER 3

3.2. Explain whether each of the following statements describes a change in demand or a change in quantity demanded, and specify whether each change represents an increase or a decrease.

- Julia believes the price of tires will rise next month, so she purchases a set of four for her pickup truck today.
- After an article is published asserting that eating quinoa causes hair loss, sales of quinoa drop by 75 percent.
- The Oink-N-Chew company experiences a significant decline in sales when it doubles the price of its bacon-flavored bubblegum.
- An increase in the federal minimum wage results in a decline in sales of fast food.
- An unexpected decrease in the price of peanut butter results in an increase in banana sales.

4.1. The market for fitness trackers is made up of five firms, and the data in the following table represents each firm's quantity supplied at various prices. Fill in the column for the quantity supplied in the market, and draw a supply graph showing the market data.

Quantity supplied by:

Price	Firm A	Firm B	Firm C	Firm D	Firm E	Market
\$ 25	5	3	2	0	5	
50	7	5	5	3	6	
75	9	7	8	6	7	
100	11	10	11	9	8	

5.7. Suppose the market demand for pizza is given by $Q_d = 300 - 20P$ and the market supply for pizza is given by $Q_s = 20P - 100$, where P = price (per pizza).

- Graph the supply and demand schedules for pizza using \$5 through \$15 as the value of P .
- In equilibrium, how many pizzas would be sold and at what price?
- What would happen if suppliers set the price of pizza at \$15? Explain the market adjustment process.
- Suppose the price of hamburgers, a substitute for pizza, doubles. This leads to a doubling of the demand for pizza. (At each price, consumers demand twice as much pizza as before.) Write the equation for the new market demand for pizza.
- Find the new equilibrium price and quantity of pizza.

CHAPTER 6

3.6. Gorgonzola is a small island nation with a simple economy that produces only six goods: sugar cane, yo-yos, rum, peanuts, harmonicas, and peanut butter. Assume that one-quarter of all the sugar cane is used to produce rum and one-half of all the peanuts are used to produce peanut butter.

- Use the production and price information in the table to calculate nominal GDP for 2018.
- Use the production and price information in the table to calculate real GDP for 2016, 2017, and 2018 using 2016 as the base year. What is the growth rate of real GDP from 2016 to 2017 and from 2017 to 2018?
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	2016		2017		2018	
Product	Quantity	Price	Quantity	Price	Quantity	Price
Sugar cane	240	\$0.80	240	\$1.00	300	\$1.15
Yo-yos	600	2.50	700	3.00	750	4.00
Rum	150	10.00	160	12.00	180	15.00
Peanuts	500	2.00	450	2.50	450	2.50
Harmonicas	75	25.00	75	30.00	85	30.00
Peanut butter	100	4.50	85	4.50	85	5.00

CHAPTER 7

1.10. In each of the following cases, classify the person as cyclically unemployed, structurally unemployed, frictionally unemployed, or not in the labor force. Explain your answers.

- Samuel quit his job as a bank teller to work full-time on his master's degree.
- Charmaine lost her job as a customer service representative when her employer outsourced the work to India.
- Lucy just graduated from law school and is deciding which job offer she will accept.
- Carlos got laid off from his job as a limo driver six months ago and gave up looking for a new job two months later.
- Byron quit his teaching job in Atlanta six weeks ago to look for a higher-paying job in Miami. He is still looking for a job.
- Arlisha lost job as a restaurant manager due to a recession.

2.3. The CPI is a fixed-weight index. It compares the price of a fixed bundle of goods in one year with the price of same bundle of goods in some base year. Calculate the price of a bundle containing 50 units of good *X*, 125 units of good *Y*, and 100 units of good *Z* in 2016, 2017, and 2018. Convert the results into an index by dividing each bundle price figure by the bundle price in 2016 and multiplying by 100. Calculate the percentage change in your index between 2016 and 2017 and again between 2017 and 2018. Was there inflation between 2017 and 2018?

	Quantity	2016	2017	2018
Good	Consumed	Prices	Prices	Prices
X	50	\$2.00	\$1.50	\$2.00
Y	125	3.00	3.00	2.50
Z	100	0.75	1.25	1.50

2.6. The CPI is 120 in year 1 and 150 in year 2. All inflation is anticipated. If the Gringotts Bank charges an interest rate of 30 percent in year 2, what is the bank's real interest rate?